Team-0510 SLU MAD Team 8

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SkillBytes: App Proposal

A mobile app for learning practical tech skills in small, quick chunks

Practical Learning Platform End to End Blueprint

1. What is SkillBytes?

SkillBytes is a mobile app for learning practical tech skills like coding, networking, and electronics. Instead of sitting through hour-long videos, learners work through short modules of about 5 to 10 minutes each. That makes it easy to learn on a bus ride, between classes, or during a break.

There are two kinds of users. **Learners** find courses, enroll, and track their progress. **Admins** create and manage the courses.

Our goal: make hands-on tech learning easy to start and easy to stick with.

What we want to achieve:

- Give beginners clear learning paths in practical tech topics.
- Let people learn in short sessions and see how far they have come.
- Give admins a simple way to publish and update content.
- Build a clean, easy-to-use mobile app in Flutter.

2. The Problem

A lot of people want to pick up tech skills but get stuck along the way:

- Most courses are long and need a big block of free time.
- Beginners don't know where to start, so they jump between random tutorials.
- Electronics and networking are hard to find in a phone-friendly format.
- Without visible progress, it's easy to lose motivation and quit.

SkillBytes tackles this with clear tracks, small modules, and a progress bar that shows what is done and what comes next.

3. Who It's For

Learners

Students, beginners exploring tech careers, hobbyists, and working professionals who want to upskill a little at a time. They need clear learning paths, short lessons, easy browsing, and a way to see what they have finished.

Admins

Instructors, subject experts, or whoever manages the platform. They need a simple way to add and edit courses and modules, organize content into tracks, and see how many learners have enrolled.

4. How Content Is Organized

Everything is split into four levels, so every lesson stays small and every course has a clear scope.

Level	What it is	Example
Track	A broad skill area	Electronics
Course	A focused topic inside a track	Arduino Basics
Module	A group of related lessons	Working with LEDs
Lesson	A 5 to 10 minute piece of reading, steps, or a quiz	Wiring your first LED

5. Planned Features

For learners

- **Sign up and login:** create an account and sign in.
- **Home screen:** a "Continue Learning" section, recommended courses, and track categories.
- **Course listing:** browse all courses, with search and category filters (Coding, Networking, Electronics).
- **Course details:** description, difficulty, estimated time, the list of modules, and an Enroll button.
- **Enrollment:** enrolled courses show up under "My Courses".
- **Lessons:** read short lessons and mark them complete.
- **Progress tracking:** a progress bar for each course and a list of completed modules.
- **Profile:** account details, enrolled courses, completed modules, and logout.

For admins

- **Admin login:** separate access for admin accounts.
- **Dashboard:** quick numbers for total courses, modules, and enrolled learners.
- **Course management:** add, edit, and delete courses.
- **Module management:** add, reorder, edit, and remove modules inside a
- course. **Learner overview:** see who is enrolled in each course.

6. User Journeys

Journey 1: A new learner starts a course

1. A first-year student opens SkillBytes and signs up with their email.
2. On the Home screen they tap the **Electronics** track.
3. They scroll the course list and open **Arduino Basics**.
4. The details page shows six short modules, so they tap **Enroll**.
5. They start "Working with LEDs" and finish the lesson in about 6 minutes.
6. They check their Profile and see their progress bar at 15 percent.

Journey 2: A returning learner picks up where they left off

1. A returning learner logs in on their commute.
2. Home shows "Continue Learning: Networking Fundamentals, Module 3".
3. They tap it and land right where they stopped.
4. They finish the module, and the app marks it done and suggests the next one.

Journey 3: An admin adds a new course

1. An admin logs in and lands on the Admin Dashboard.
2. The admin taps **Add Course** and fills in the title, track, difficulty, and description.
3. The admin adds three modules with short lessons and saves.
4. The new course now appears in the learners' Course Listing.
5. A few days later the admin opens it and checks how many learners have enrolled.

7. Tools We Plan to Use

Flutter (Dart) to build the mobile app

Git and GitHub for version control

Figma or Whimsical for wireframes

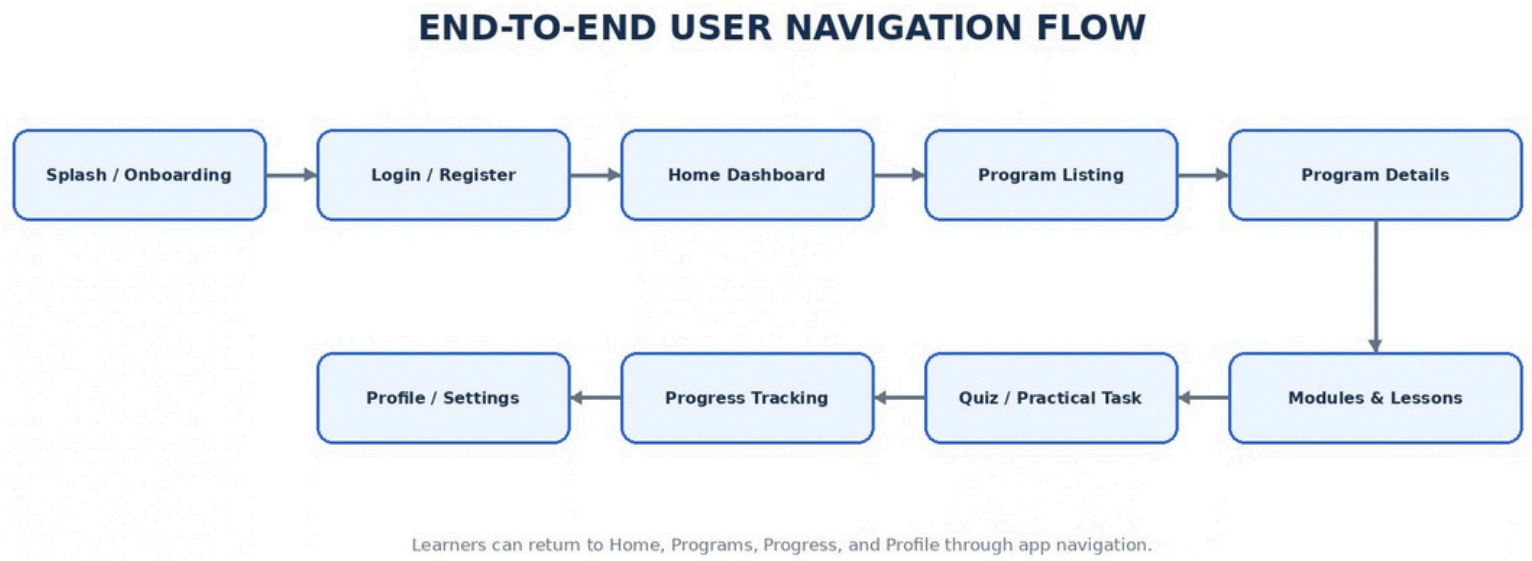
Mock data first, with a backend like Firebase as a possible upgrade later

8. Navigation Flow

Learner: Login or Sign up, then Home, then Course Listing, then Course Details, then Lesson. Profile is always one tap away in the bottom bar.

Admin: Login, then Dashboard, then Manage Courses, then Add or Edit Course, then Manage Modules.

The learner app uses a bottom navigation bar with three tabs: **Home**, **Courses**, and **Profile**

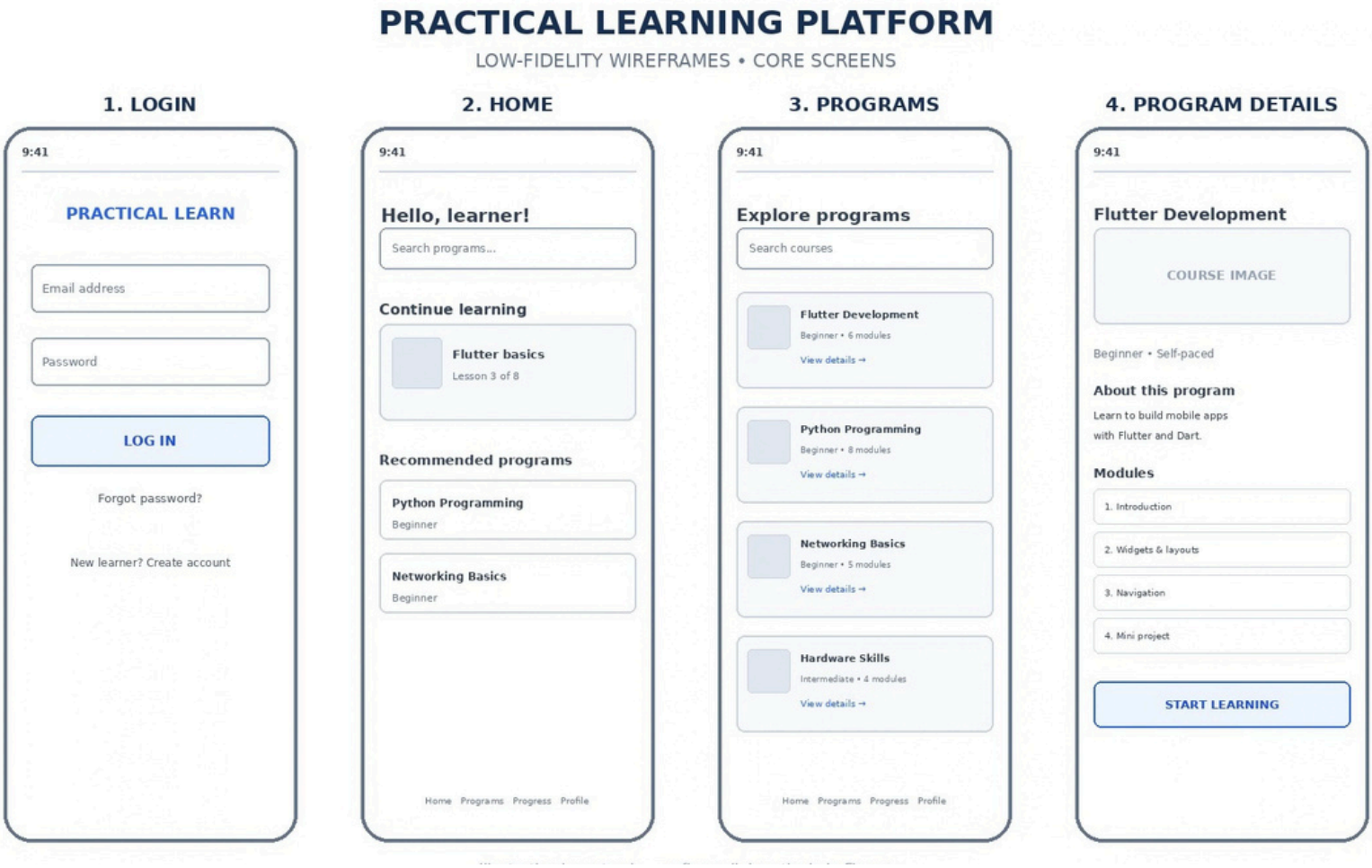


9. Wireframe Screens

For Week 1 we are sketching four screens:

1. Login and Sign up
2. Home
3. Programs
4. Program Details

These are low-fidelity sketches, not final UI designs.
We plan to add a Progress bar too



10. Scope

Included in the first version: login and sign up screens, home, course listing and details, enrollment, progress tracking, profile, and a basic admin panel.

Maybe later: offline lessons, quizzes with scores, certificates, reminder notifications, interactive circuit simulations, and a community discussion space.

11. Code Repository

GitHub repository: **GitHub Repo SkillBytes**

The repo holds the starter Flutter project and a README that explains the project vision, objectives, and navigation flow.

12. Wrapping Up

SkillBytes is a simple idea with a clear purpose: make practical tech skills easier to learn, one small step at a time. With a clean content structure and straightforward navigation for both learners and admins, it gives us a solid base to build on in the coming weeks.